Dyck Skeleton String Decompositions

1 Overview

This note studies string decompositions inside fixed deficit layers. A string is a sequence of paths along which area increases by 1, dinv decreases by 1, and deficit remains fixed. The guiding example is Hawkes's low-deficit Dyck skeleton formula, where special skeletons start strings whose lower halves cover the corresponding deficit layers.

The main conjectural extension considered here is the case $r = \tau s + 1$. For this family we define τ-Dyck sequences, a τ-analogue of dinv, special skeletons, and a conjectural lower-half string formula. A full coefficient formula requires either deficit-layer q, t-symmetry or a direct full coefficient-dictionary identity.

The final sections describe finite computational checks. These checks support the conjectural formula in selected ranges and also test a separate general-slope diagnostic called NRCM.

All coefficient polynomials in this note use the convention $q^{\mathrm{area}} t^{\mathrm{dinv}}$. Thus the lower half of a fixed deficit layer is the part with area $\leq$ dinv.

The rest of the note first introduces the mathematical sequences, paths, and claims and then describes the finite checks attached to them.

2 Common Vocabulary

For a fixed total degree M, the deficit of a sequence or path X is

$$\mathrm{defc}(X) = M - \mathrm{area}(X) - \mathrm{dinv}(X).$$

A string of fixed deficit e is a sequence

$$X_0, X_1, \ldots, X_m$$

such that $\mathrm{area}(X_{i+1}) = \mathrm{area}(X_i) + 1$, $\mathrm{dinv}(X_{i+1}) = \mathrm{dinv}(X_i) - 1$, and $\mathrm{defc}(X_i) = e$ for every i. Thus every term of the string has total degree $M - e$.

The lower half of deficit e means the paths with

$$\mathrm{area} \leq \left\lfloor \frac{M - e}{2} \right\rfloor.$$

A lower-half decomposition is a disjoint cover of this lower-half set by the initial segments of strings.

The remaining sections use this vocabulary in three settings: the classical Dyck setting, the $r = \tau s + 1$ rational setting, and the NRCM diagnostic. The mathematical definitions are stated first. The scripts and finite checks are described separately afterward.

3 The Classical Model

An ordinary Dyck sequence of length n is an integer sequence

$$x = (x_0, \ldots, x_{n-1})$$

with

$$x_0 = 0, \qquad x_i \geq 0, \qquad x_{i+1} \leq x_i + 1.$$

Its area is

$$\operatorname{area}(x) = \sum_{i=0}^{n-1} x_i,$$

and the classical dinv statistic in this notation is

$$\operatorname{dinv}(x) = \#\{(i,j) : 0 \le i < j < n,\ x_i = x_j \text{ or } x_i = x_j + 1\}.$$

Here $M = \binom{n}{2}$, so

$$\operatorname{defc}(x) = \binom{n}{2} - \operatorname{area}(x) - \operatorname{dinv}(x).$$

The skeleton terminology in this section is imported from Section 5 of [1]. In Hawkes's setup, extractable entries are the entries removed by the recursive down move, full Dyck skeletons are the terminal Dyck sequences with no extractable entry, and special Dyck skeletons are the permitted string starts after Hawkes's special attachment convention. The up and down moves then generate the strings.

The classical theorem used as the template here is Hawkes's low-deficit skeleton formula. All formulas in this note use the convention $q^{\text{area}}t^{\text{dinv}}$:

$$C_n(q,t)\big|_{\binom{n}{2}-2n+8 \le \deg_{q,t} \le \binom{n}{2}} = \sum_{\substack{S \text{ special Dyck skeleton of length } n \\ \operatorname{defc}(S) \le 2n-8}} \frac{q^{\operatorname{area}(S)}t^{\operatorname{dinv}(S)+1} - q^{\operatorname{dinv}(S)+1}t^{\operatorname{area}(S)}}{t-q}.$$

The rational expression is a finite interval contribution: if $a = \operatorname{area}(S)$, $\nu = \operatorname{dinv}(S)$, and $e = \operatorname{defc}(S)$, then in the low-deficit range of the theorem one has $a \le \nu$, and

$$\frac{q^a t^{\nu+1} - q^{\nu+1}t^a}{t-q} = \sum_{i=a}^{\nu} q^i t^{M-e-i}.$$

Hawkes proves that these strings cover the lower half of every deficit layer with $\operatorname{defc} \le 2n-8$. Equivalently, their lower-half contributions are truncated at the midpoint:

$$\sum_{i=a}^{\lfloor (M-e)/2 \rfloor} q^i t^{M-e-i}.$$

The upper half is then supplied by the symmetry $C_n(q,t) = C_n(t,q)$.

4 The τ-Dyck Model for $r = \tau s + 1$

Fix integers $s \ge 1$ and $\tau > 1$, and put $r = \tau s + 1$. For the conjectural model in this note, the ordinary Dyck growth bound $+1$ is replaced by $+\tau$.

A τ-Dyck sequence of length s is an integer sequence

$$x = (x_1, \ldots, x_s), \qquad x_1 = 0, \qquad x_i \ge 0, \qquad x_{i+1} \le x_i + \tau.$$

These inequalities imply

$$0 \le x_i \le \tau(i-1).$$

Thus there are only finitely many τ-Dyck sequences of each length.

The area statistic is

$$\operatorname{area}(x) = \sum_{i=1}^{s} x_i.$$

The dinv statistic used in this note is the following τ-analogue:

$$\operatorname{dinv}(x) = \sum_{1 \le i < j \le s} \begin{cases} \max(0, x_i + \tau - x_j), & x_i \le x_j, \\ \max(0, x_j + 1 + \tau - x_i), & x_i > x_j. \end{cases}$$

The total degree is

$$M = \tau\binom{s}{2},$$

and

$$\mathrm{defc}(x) = M - \mathrm{area}(x) - \mathrm{dinv}(x).$$

For a deficit cutoff B, define the aggregate polynomial

$$\mathcal{C}_{\tau,s}^{\le B}(q,t) = \sum_{\substack{x \text{ is a } \tau\text{-Dyck sequence} \\ 0 \le \mathrm{defc}(x) \le B}} q^{\mathrm{area}(x)} t^{\mathrm{dinv}(x)}.$$

The conjectural bound is

$$B = (s-2)(\tau+1) - 4.$$

For comparison with the position-coordinate notation used later in the NRCM diagnostic, when $r = \tau s+1$ one has $\lfloor ri/s \rfloor = \tau i$ for $0 \le i < s$, so $\Delta_i = \tau$ for $0 \le i < s-1$. In Q-coordinates, path-validity gives

$$Q_{i+1} \le Q_i + \tau \qquad (0 \le i < s-1).$$

Identifying $x_{i+1} = Q_i$ gives exactly the τ-Dyck growth bound $x_{i+2} \le x_{i+1} + \tau$.

5 Rational Special Skeletons

For this conjectural model, all skeletons are τ-Dyck sequences of the fixed length s. Deletion is used only to define extractability and the recursive map operations.

Let $x = (x_1, \ldots, x_s)$ be a τ-Dyck sequence. An entry x_j is extractable if:

$$x_j > 0,$$
$$\#\{i < j : \max(0, x_j - \tau) \le x_i < x_j\} = 1,$$

and deleting x_j leaves a τ-Dyck sequence. The endpoint conventions are explicit:

- $j = 1$ cannot occur because $x_1 = 0$;
- if $1 < j < s$, deletion is equivalent to the splice inequality

$$x_{j+1} \le x_{j-1} + \tau;$$

- if $j = s$, there is no extra splice inequality.

The sequence after deletion is

$$(x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_s).$$

The recursive moves use left-to-right extraction order: the first extractable entry is the one removed.

A full skeleton is a τ-Dyck sequence of length s with no extractable entry. For $s \ge 4$, one full skeleton is excluded from the starting set:

$$(0, 0, 1, \underbrace{0, \ldots, 0}_{s-4 \text{ zeros}}, \tau).$$

For $s = 4$ this is $(0, 0, 1, \tau)$. For $s < 4$, there is no excluded full skeleton and no special upward attachment. A special skeleton is a full skeleton other than this excluded sequence.

The excluded full skeleton is not removed from the strings. It is attached to the string beginning at

$$(0, \ldots, 0, \tau)$$

by the special upward move

$$(0, \ldots, 0, \tau) \longmapsto (0, 0, 1, \underbrace{0, \ldots, 0}_{s-4 \text{ zeros}}, \tau) \qquad (s \ge 4).$$

6 Conjectural Lower-Half Formula and Symmetry Completion

Conjecture 1 (Special-skeleton lower-half formula for $r = \tau s + 1$)**.** *Fix $s \geq 1$ and $\tau > 1$, and set $M = \tau\binom{s}{2}$ and*

$$B = (s-2)(\tau+1) - 4.$$

For $0 \leq d \leq B$, put

$$L_d = \left\lfloor \frac{M-d}{2} \right\rfloor.$$

If $B \geq 0$, the lower-area part of each deficit layer $d \leq B$ is obtained by summing over all special skeletons z of length s with $\mathrm{defc}(z) = d$. These skeletons are lower-half starts. Write

$$a = \mathrm{area}(z), \qquad \nu = \mathrm{dinv}(z), \qquad d = \mathrm{defc}(z).$$

Then z contributes the lower-half interval ending at the cutoff L_d:

$$\sum_{i=a}^{L_d} q^i t^{M-d-i}.$$

The asserted identity is

$$\sum_{\substack{x \text{ is a } \tau\text{-Dyck sequence} \\ \mathrm{defc}(x)=d,\ \mathrm{area}(x) \leq L_d}} q^{\mathrm{area}(x)} t^{\mathrm{dinv}(x)} = \sum_{\substack{z \text{ special skeleton} \\ \mathrm{defc}(z)=d}} \sum_{i=\mathrm{area}(z)}^{L_d} q^i t^{M-d-i}.$$

The full aggregate polynomial $\mathcal{C}^{\leq B}_{\tau,s}(q,t)$ follows from this lower-half statement only with an additional symmetry input. The needed symmetry is deficit-layer symmetry: for each $0 \leq d \leq B$,

$$\sum_{\mathrm{defc}(x)=d} q^{\mathrm{area}(x)} t^{\mathrm{dinv}(x)} = \sum_{\mathrm{defc}(x)=d} q^{\mathrm{dinv}(x)} t^{\mathrm{area}(x)},$$

where the sums run over τ-Dyck sequences of length s. Equivalently, one may prove the full coefficient dictionary directly. Without one of these two inputs, the displayed conjecture is only a lower-half string-start formula.

A useful algebraic packaging of a full string interval is the rational expression

$$\frac{q^a t^{\nu+1} - q^{\nu+1} t^a}{t-q}.$$

When $a \leq \nu$, this expands as the full interval

$$\sum_{i=a}^{\nu} q^i t^{M-d-i}.$$

If $a > \nu$, the same expression expands with the opposite sign over the open interval between ν and a. That signed expansion is algebraic bookkeeping, not the lower-half string interpretation above.

7 Computations for the Conjecture

The relevant scripts are

`code/check_r1mod_skeleton_strings.py` and `code/run_official_r1mod_checks.py`.

Some script options and output fields use the legacy name `defect`. The relevant names here are

`--max-defect` and `initial_passing_defect_range`.

In the mathematical exposition these refer to deficit.

The single-case command

```
python code/check_r1mod_skeleton_strings.py --tau T --s S
```

uses $B = (S-2)(T+1)-4$ unless `--max-defect` is supplied. It performs a finite formula check by comparing two coefficient dictionaries. The direct dictionary is obtained by exhaustive τ-Dyck sequence enumeration in the retained deficit range, with pruning only when an exact suffix bound proves that the remaining suffix cannot reach the required total area + dinv. The predicted dictionary in the script is obtained from the signed rational expressions above, so this check is a direct full-dictionary identity rather than only a lower-half comparison. In the exposition here, the signed terms should be read as algebraic bookkeeping; the string picture is the lower-half formula plus deficit-layer symmetry or direct full-dictionary agreement. The formula status is `PASS` iff the two dictionaries are identical. If the requested bound is negative, there is no nonnegative deficit layer to test.

For $S \geq 5$, the same script also attempts a lower-half map check. The implemented local cases are named East3, East5, West3, and West5 in the code. This note treats those labels as implementation-defined local moves. The special-skeleton map check tests the implemented up and down moves: preservation of deficit, area change by 1, inverse behavior on checked steps, and disjoint coverage of the lower-half target. A map status of `PASS` means that the checked lower-half strings cover the target through the supported local cases. A status of `PARTIAL` means that no contradiction was found before the implemented moves reached the code label `unsupported_level_7`. For $s \geq 4$, this map check also includes the special upward attachment from $(0, \ldots, 0, \tau)$ to the excluded full skeleton.

The lower-cutoff roots also have a partition-indexed count. The accompanying code verifies the following refined form: for defc $\leq (s-2)\tau - 1$, special skeleton roots of deficit d and area a are counted by partitions of size d with a parts and with all parts at most $s-2$. Equivalently, after conjugating partitions this can be phrased using partitions of length at most $s-2$, but then the root area records the largest part of the conjugate partition rather than its number of parts.

The batch runner is configured to run the following finite grid. Since no run log is included here, this table records the intended finite test range, not independent evidence that each listed command has been freshly run.

τ	formula cases	formula+map cases
2	$1 \leq s \leq 14$	$5 \leq s \leq 14$
3	$1 \leq s \leq 12$	$5 \leq s \leq 12$
4	$1 \leq s \leq 10$	$5 \leq s \leq 10$
5	$1 \leq s \leq 9$	$5 \leq s \leq 9$

The reproducible command is

```
python code/run_official_r1mod_checks.py.
```

For example, at $(\tau, s) = (2, 15)$ the conjectural bound is exactly

$$(15-2)(2+1) - 4 = 35.$$

The command

```
python code/check_r1mod_skeleton_strings.py --tau 2 --s 15 --report-level7
```

is designed to test the formula through precisely the conjectural bound defc ≤ 35 and reports any level-7 obstructions encountered by the partial map.

8 The Naive Rational Cyclic Map

The NRCM diagnostic applies to coprime positive integers r and s with $s > 1$. Let

$$H_i = \left\lfloor \frac{ri}{s} \right\rfloor, \qquad L_i \in \{0, \ldots, s-1\}, \quad L_i \equiv ri \pmod{s}, \qquad \Delta_i = H_{i+1} - H_i$$

for $0 \le i < s$, where Δ_i is used only for $0 \le i < s-1$. A position-coordinate path is an integer sequence

$$Q = (Q_0, \ldots, Q_{s-1})$$

with

$$Q_0 = 0, \qquad 0 \le Q_i \le H_i.$$

It is path-valid when the path heights

$$P_i = H_i - Q_i$$

satisfy

$$P_0 \le P_1 \le \cdots \le P_{s-1}.$$

The diagnostic area is

$$\mathrm{area}(Q) = \sum_{i=0}^{s-1} Q_i,$$

and $M = \sum_i H_i$. The diagnostic deficit is defined by

$$\mathrm{defc}(Q) = \sum_{1 \le i < j < s} \delta_{ij}(Q),$$

where the summand is as follows. Put

$$u_{ij} = |Q_i - Q_j|.$$

If $Q_i \ne Q_j$ and the comparisons $Q_i > Q_j$ and $L_i > L_j$ have opposite truth values, replace u_{ij} by $u_{ij} - 1$; then replace it by $\max(u_{ij}, 0)$. Define

$$v_{ij} = \begin{cases} \Delta_i - (Q_{i+1} - Q_i), & Q_i > Q_j, \\ \Delta_{i-1} - (Q_i - Q_{i-1}), & Q_j > Q_i, \\ 0, & Q_i = Q_j. \end{cases}$$

Then

$$\delta_{ij}(Q) = \min(u_{ij}, v_{ij}).$$

The second quantity v_{ij} is not separately clamped in this diagnostic definition. On path-valid inputs it is nevertheless nonnegative: the two nonzero cases are exactly the adjacent inequalities $Q_{i+1} - Q_i \le \Delta_i$ and $Q_i - Q_{i-1} \le \Delta_{i-1}$. No replacement by $\max(v_{ij}, 0)$ is part of the implemented statistic. Finally,

$$\mathrm{dinv}(Q) = M - \mathrm{area}(Q) - \mathrm{defc}(Q).$$

These are the statistics used by the NRCM scripts.

For a suffix $I_k = \{k, k+1, \ldots, s-1\}$, list its columns in increasing label order L_i. The candidate $T_k(Q)$ cyclically moves the Q-value from each listed column to the next listed column and moves the last listed value to the first listed column after adding 1. Columns outside I_k are unchanged. The strict NRCM tries $k = 1, 2, \ldots, s-1$, stops at the first k for which $T_k(Q)$ satisfies the capacity inequalities

$$0 \le T_k(Q)_i \le H_i,$$

and defines

$$\mathrm{NRCM}(Q) = T_k(Q)$$

only if this first capacity-valid candidate is also path-valid. If no suffix is capacity-valid, or if the first capacity-valid suffix is not path-valid, then $\mathrm{NRCM}(Q)$ is undefined.

Here is a small successful move. For slope $5/3$,

$$H = (0, 1, 3), \qquad L = (0, 2, 1).$$

Let $Q = (0, 1, 1)$. For $k = 1$, the suffix columns $\{1, 2\}$ appear in label order $2, 1$. Thus $Q_2 = 1$ moves to column 1, and $Q_1 = 1$ wraps to column 2 as 2, giving

$$T_1(Q) = (0, 1, 2).$$

This candidate is capacity-valid and path-valid.

The failure mode in the definition is also important. For slope $4/3$,

$$H = (0, 1, 2), \qquad L = (0, 1, 2),$$

and $Q = (0, 0, 1)$. The first capacity-valid suffix is $k = 2$, giving

$$T_2(Q) = (0, 0, 2).$$

Its path heights are $(0, 1, 0)$, which are not nondecreasing. Strict NRCM is therefore undefined on this Q. In this example $k = 2$ is already the final suffix, so there is no valid NRCM value.

9 NRCM Lower-Half Diagnostics

The NRCM scripts are

`code/check_nrcm_lower_half.py` and `code/check_nrcm_domain.py`.

The lower-half script checks, in finite deficit layers, whether every below-midline source has a defined NRCM image, whether the image has the same diagnostic deficit and area one larger, whether the image remains in the allowed target, and whether the map is injective on the checked sources. The domain script is narrower: it checks only definedness on below-midline sources.

By definition, every defined strict NRCM move raises area by 1: the suffix values are cyclically permuted and the wrapped value is increased by 1. The proof-status issue concerns deficit preservation. A proof that defined strict NRCM preserves the diagnostic deficit exists outside this note and has been partially human-verified, but it is not part of the present argument. The finite diagnostics below locate the deficit layers where the directed-chain picture can be checked directly.

Proposition 1 (Conditional NRCM chain consequence)**.** *Fix a slope r/s, a deficit e, and $M = \sum_i H_i$. Let*

$$\mathcal{L}_e = \{Q : \ Q \textit{ is valid}, \ \mathrm{defc}(Q) = e, \ \mathrm{area}(Q) \le \lfloor (M - e)/2 \rfloor\}$$

be the checked lower-half target, including midpoint targets when $(M - e)/2$ is an integer. Suppose strict NRCM is defined on every valid path of deficit e with

$$\mathrm{area}(Q) < \frac{M - e}{2},$$

so no move is required from a midpoint target. Suppose the checked moves preserve deficit, raise area by 1, land in $\mathcal{L}_e$, and are injective on those sources. Suppose also that every path in $\mathcal{L}_e$ which is not declared NRCM-minimal has a checked predecessor in the same deficit layer. Then $\mathcal{L}_e$ is covered by directed NRCM chains. The chain starts are the NRCM-minimal paths, meaning valid paths of deficit e that are not the NRCM image of any lower-area path in the same checked layer.

To turn such lower-half strings into a full coefficient formula, one additionally needs equality of the predicted interval dictionary with the full direct coefficient dictionary, or a separately proved q, t-symmetry for the homogeneous deficit layer being considered.

The output field

```
initial_passing_defect_range:  0..D
```

means that every deficit layer $0, 1, \ldots, D$, inclusive, passed the specific diagnostic run. For the lower-half script this includes definedness, deficit preservation, area increase, target membership, and injectivity. For the domain script it includes only definedness. The notes accompanying these values record tentative patterns; claims that use those patterns should include the corresponding command and run output.

10 Status of Claims

claim	status
classical Dyck skeleton formula	proved by Hawkes, Theorem 5.35 of [1], for $n \geq 4$ and $\mathrm{defc} \leq 2n-8$
$r = \tau s + 1$ special-skeleton lower-half formula	conjectural; finite exhaustive coefficient checks are implemented for selected (τ, s), and full recovery also needs deficit-layer symmetry or direct full-dictionary agreement
East3/East5/West3/West5 lower-half map	finite computational diagnostic using the local moves named in the code; the implemented map stops when an unsupported level-7 case is reached
NRCM deficit preservation	conditional in this note outside the finite runs; defined NRCM moves raise area by 1 by construction, while preservation of diagnostic deficit is the separate proof-status claim

The NRCM material is diagnostic rather than a complete theorem in this note. The definitions, examples, and finite checks are included because they test the same directed-chain picture for general coprime slopes. Claims depending on NRCM deficit preservation should be read as conditional unless accompanied by a separate proof or by the relevant finite run output.

References

[1] Graham Hawkes, *Dyck Symmetric Functions and Applications to q,t-Catalan Polynomials*, arXiv:2605.13003, 2026.
`https://arxiv.org/abs/2605.13003`